

Day 7: STTP on Ethical Hacking and Cyber Forensics @IIITK

Digital Forensics Data Acquisition and Analysis
and Memory Forensics to Detect Malicious Executables

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June 17, 2025

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1 Introduction to Digital Forensics

Digital Forensics is a branch of forensic science for identifying, acquiring, processing, analyzing, and reporting on electronically stored data in a forensically sound manner.

- **Key Principles:** Analysis, Validation (using hash functions), Reporting, Chain of Custody, and validated Tools.
- *Hash is a one-way function; data tampering changes hash value. Two hashes are recommended.*
- *Open source tools may not be accepted in some courts.*

1.1 Digital Forensic Investigation Process

The process involves:

1. **Evidence Collection**
2. **Evidence Preservation**
3. **Evidence Analysis**
4. **Evidence Presentation**

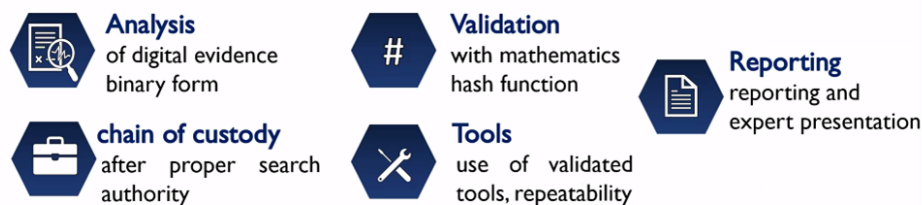


Figure 1: Digital Forensic Investigation Process

1.2 Types of Digital Evidence

Evidence can be found from web history, emails, mobile devices (location, calls), and smart devices (rings, watches, vehicles, locks, lights, smart assistants).

2 Data Acquisition in Digital Forensics

Data acquisition creates a forensic image (a copy of data) for analysis.

2.1 Data Acquisition Considerations

- **Physical vs. Logical Acquisition:** Physical (bit-by-bit copy, most complete) and Logical (active files only).
- **Data Integrity:** Hashing ensures acquired data is correct and verifiable.

2.2 Write Blocker

A write blocker is a hardware/software component that prevents data alteration on a suspect's storage device, crucial for preserving evidence integrity.



Figure 2: Example of a Hardware Write Blocker

2.3 Data Acquisition Methodology

Methodology depends on whether the computer is on or off. Volatile data (RAM) is acquired first if the system is on.

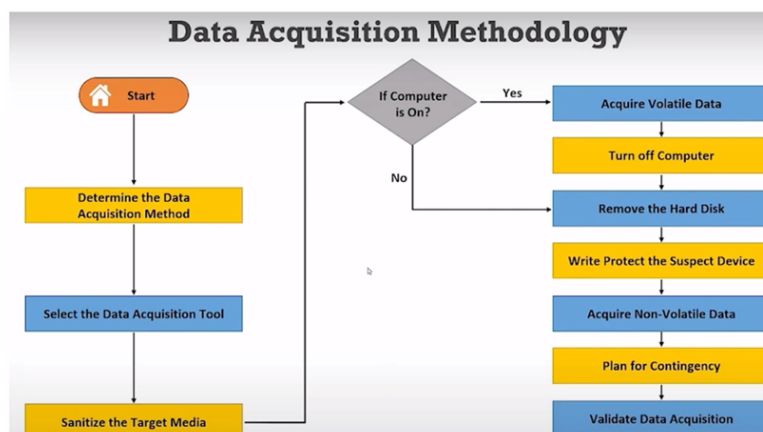


Figure 3: Data Acquisition Methodology Flowchart

2.4 Mobile Phone Acquisition

Mobile acquisition can be Logical (OS accessible), File System (files, hidden files), or Physical (bit-for-bit, includes deleted data). Physical acquisition often uses hardware methods like JTAG or Chip Off. *Hashing is generally not valid for dynamic mobile data (e.g., alarms, messages).*

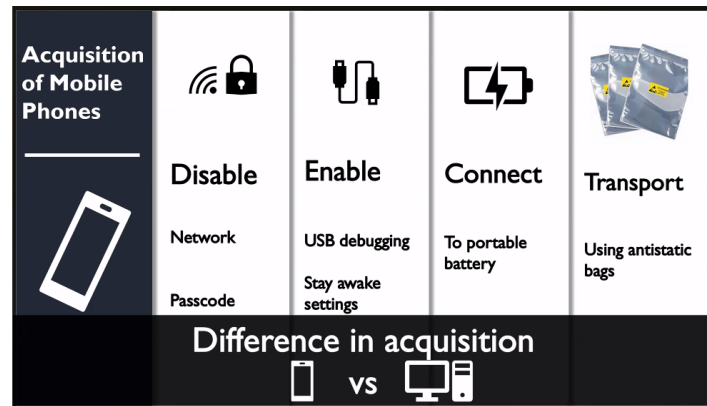


Figure 4: Mobile Phone Data Acquisition Types

3 Digital Forensic Analysis Tools

Various tools are used, including: Autopsy, EnCase, Oxygen Forensics, Magnet Axion, FTK Imager, X-Ways, Cellebrite, Belkasoft, Volatility (Memory Forensics), and Wireshark (Network Forensics).



Figure 5: Digital Forensic Analysis Tools

3.1 FTK Imager

FTK Imager is a free tool for creating disk images and capturing memory.

3.1.1 Creating a Disk Image

1. Go to File → Create Disk Image.
2. Select source type (Physical Drive, Logical Drive, etc.).
3. Specify destination path and image format (Raw, E01, AFF).
4. Add case details. FTK Imager computes MD5/SHA1 hashes automatically.

3.1.2 Capturing Memory

1. Go to File → Capture Memory.
2. Choose destination and filename (e.g., `memdump.mem`). Optionally include pagefile.
3. *Memory dumps do not always require hash values.*
4. RAM capture bypasses most encryption.

4 Memory Forensics for Detecting Malicious Executables

Memory forensics analyzes RAM to find malicious activity.

4.1 Why RAM is Important

RAM contains executed programs, opened web pages, decrypted content, and network traffic. Fileless malware resides only in RAM and may evade antivirus, detected only via RAM dump.

4.2 Forensic Value

It helps identify suspicious processes, network activity, malware, and persistence. "Malware can hide but it has to run," leaving traces in memory.

4.3 Volatility Tool

Volatility is an open-source Python tool for analyzing memory images (Windows, Linux, macOS). **Triage Process:** 1. Collect running processes. 2. Check for deviations. 3. Identify leads.

4.4 Volatility Plugins and Process Interpretation

- `imageinfo`: Identifies OS profile (e.g., WinXPSP2x86).

```
volatility -f cridex.vmem imageinfo
```

- `pslist`: Lists running processes.

```
volatility -f cridex.vmem --profile=WinXPSP2x86 pslist
```

- **psscan:** Finds hidden/terminated processes.
- **connscan:** Checks network connections.

```
volatility -f cridex.vmem connscan
```

- **procdump/memdump:** Dumps process executable/memory for analysis.

```
volatility -f cridex.vmem procdump -p 1640 --dump-dir .
```

- **malfind:** Finds injected code (e.g., 'PAGE_EXECUTE_READWRITE' permissions).

```
volatility -f cridex.vmem malfind -p 1640
```

- **yarascan:** Scans memory with YARA rules to identify malware.

```
volatility -f stuxnet.vmem yarascan -y malware_rules.yar
```

4.4.1 Interpreting Processes (Triage)

Look for deviations from expected behavior in:

- **Process Path:** Running location.
- **Parent-Child Relationships:** Unexpected parents.
- **Singleton:** Processes that should have only one instance.
- **Account:** User account running the process.
- **Start Time:** Boot time processes.
- **Misspellings/Fake Names:** Common malware masquerading.

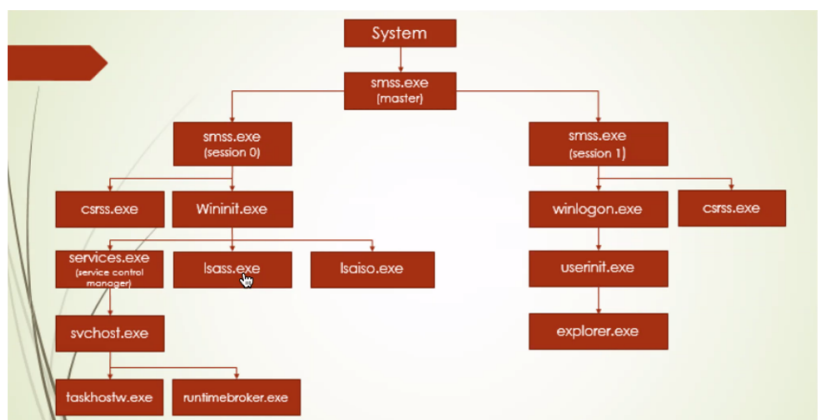


Figure 6: Windows Core Process Hierarchy (Example)

Process	Parent	Path	Singleton	Account	Start time
System	none	None	Yes	Local System	boot
SMS.EXE	SYSTEM	System32\smss.exe	No	Local System	boot
wininit.exe	none	System32\wininit.exe	Yes	Local System	boot
taskhost.exe	services.exe	System32\taskhost.exe	No	many	varies
lsass.exe	wininit.exe	System32\lsass.exe	Yes	Local System	boot
winlogon.exe	none	System32\winlogon.exe	No	Local System	varies
ieexplore.exe	explorer.exe	\Program Files\Internet Explorer\explore.exe	No	Local Users	varies
explorer.exe	userinit.exe	SystemRoot%\explorer.exe	No	Local Users	varies
lsim.exe	wininit.exe	\System32\lsim.exe	Yes	Local System	boot
svchost.exe	services.exe	\System32\svchost.exe	No	Local System, Network Service, or Local Service	boot, some after boot
services.exe	wininit.exe	\System32\services.exe	Yes	Local System	boot
csrss.exe	none	\System32\csrss.exe	No	Local System	boot

Figure 7: Windows Core Processes Properties

5 Cridex Malware Case Study (Summary)

Analysis of a 'cridex.vmem' memory dump revealed:

- Suspicious 'reader_sl.exe' spawned by 'explorer.exe' at boot time.
- 'connscan' showed 'explorer.exe' connecting to '41.168.5.140:8080'.
- Dumping 'reader_sl.exe's memory showed banking domains and the C2 IP '41.168.5.140'.
- The dumped executable was identified as a Trojan.
- Registry analysis revealed 'KB00207877.exe' as a persistence mechanism.
- ****Conclusion****: The system was infected by a Trojan, using process hollowing for persistence and C2 communication with banking sites.

6 Challenges in Digital Forensics

- Lack of skilled professionals.
- Rapid technological changes.
- Dealing with encrypted data.
- Ensuring admissibility in court.
- Tracing cybercrime to locations (IP) rather than identities.

7 Various Digital Forensics Domains

Includes: Memory, Disk, Network, Cloud, Darkweb, Cryptocurrency, Mobile, IoT, Multimedia, Email, Malware, and Drone Forensics.

8 Learning and Contribution

1. Analyze Forensic Images (Autopsy, Registry Explorer).
2. Understand data parsing; develop tools/modules.
3. Create Scenarios, Forensic Images (Datasets, CTFs).
4. Reverse Engineer digital products (hardware/software).
5. Infect VM, dump memory, analyze with Volatility.
6. Develop and contribute Volatility Plugins.
7. Explore AI in digital forensics.